TASK 05-50-27-210-801-A
EFFECTIVITY: ALL

1. Overweight Landing - Inspection/Check

A. General

(1) An overweight landing inspection is necessary when these two conditions occur:

(a) The aircraft lands with weight above the MLW;

(b) The crew cannot tell if the limit sink rate of 360 fpm was obeyed.

(2) This procedure lets you use flight data source for the check of the loads at touchdown and compare it with the aircraft landing envelope. For this procedure, the vertical CG acceleration and the roll rate at touchdown are used.

(3) The flight data source can be either the DVDR 1, DVDR 2 or QAR. Any of them is sufficient to define if the overweight landing inspection is necessary.

NOTE: The DVDR 1 is installed in the FWD avionics compartment and the DVDR 2 is installed in the aft avionics compartment.

(4) When the aircraft is equipped with the QAR, you can use this equipment as an alternative to the DVDR 2.

NOTE: The QAR must operate in continuous mode to be used as an alternative to the DVDR 2.

(5) The operator can do the analysis of the flight operation data, but if the alternative is sending the data for an EMBRAER analysis, it is recommended that you send the QAR file.

(6) These maintenance technicians are necessary to do this task:

- 2 - Airframe (Structure)

B. References

REFERENCE	*DESIGNATION*
AMM MPP 53-04-20/400	WING-TO-FUSELAGE FAIRING FORWARD PANEL - REMOVAL/INSTALLATION
AMM MPP 53-04-30/400	WING-TO-FUSELAGE FAIRING CENTER PANEL - REMOVAL/INSTALLATION
AMM MPP 53-04-40/400	WING-TO-FUSELAGE FAIRING SIDE PANEL - REMOVAL/INSTALLATION
AMM MPP 53-04-50/400	WING-TO-FUSELAGE FAIRING AFT PANEL - REMOVAL/INSTALLATION
AMM SDS 52-11-00/1	FORWARD PASSENGER DOOR
AMM SDS 52-12-00/1	AFT PASSENGER DOOR
AMM SDS 52-31-00/1	FORWARD CARGO DOOR
AMM SDS 52-32-00/1	AFT CARGO DOOR
AMM SDS 52-41-00/1	FORWARD SERVICE DOOR
AMM SDS 52-42-00/1	AFT SERVICE DOOR

Filtered Serial Nº contained in this publication: 282, 336

(Continued)

REFERENCE	DESIGNATION
AMM TASK 05-50-03-200-801-A/600	Hard Landing And Off Temperature Envelope Landing - Inspection/Check
AMM TASK 05-50-12-200-801-A/600	High Energy Stop Condition - Inspection/Check
AMM TASK 07-10-01-500-801-A/200	Complete Aircraft Jacking - Maintenance Practices
AMM TASK 12-11-01-650-802-A/300	Fuel Tank - Pressure Defueling
AMM TASK 20-00-00-910-801-A/200	Aircraft Maintenance Safety Procedures - Standard Procedures
AMM TASK 31-31-00-970-801-A/200	DVDR System - Download Procedure with PCMCIA Card
AMM TASK 31-31-00-970-802-A/200	DVDR System - Download Procedure with Portable Unit
AMM TASK 31-32-00-470-801-A/200	QAR System - Programming
AMM TASK 32-11-03-200-802-A/600	Main-Landing-Gear Shimmy Damper - Inspection
AMM TASK 32-34-00-710-801-A/500	Emergency-Extension System (On Jacks) - Operational Test
AMM TASK 51-50-00-800-801-B/600	Aircraft Alignment Check - Inspection/Check
LMM SUBTASK 72-00-00-810-027	Overlimit Hard Landing

C. Zones and Accesses

ZONES ACCESS	LOCATION	REFERENCE
191AR	Wing-to-Fuselage Fairing FWD Panel	AMM TASK 06-41-00-800-801-A/100
191BL	Wing-to-Fuselage Fairing FWD Panel	AMM TASK 06-41-00-800-801-A/100
191CL	Wing-to-Fuselage Fairing FWD Panel	AMM TASK 06-41-00-800-801-A/100
191EL	Wing-to-Fuselage Fairing FWD Panel	AMM TASK 06-41-00-800-801-A/100

D. Overweight Landing

SUBTASK 210-010-A

(1) Refer to Figure 601. The flowchart is a summary of the necessary actions when an overweight landing occurs.

(2) When the crew reports a landing with the weight above the MLW, and they cannot tell if the sink rate at touchdown was less than 360 fpm, do the Aircraft Structure and engines Phase-I inspection immediately. If you do not find incorrect conditions during the Phase-I inspection, a fly-by of 10 FC is permitted. The fly-by of 10 FC is granted after the Phase I inspection without any discrepancy detected and it starts when the crew reports the event or when the flight data analysis shows the event. Then, you can do one of these two procedures:

NOTE: No inspections are necessary for sink rates less than 360 fpm.

1. At the end of the fly-by period, do the Aircraft Structure and engines Phase-II inspection. In this case, no flight data analysis is necessary. You must do Aircraft Structure and engines Phase-III inspection only if you find an incorrect condition during the Aircraft Structure and engines Phase-II inspection.

2. During the fly-by period, obey the instructions given in SUBTASK 210-003-A to analyze the flight data. Then, you must do the Phase-II inspection only if the data analysis shows that the loads exceeded the aircraft envelope. The Phase-III inspection is applicable only if you find an incorrect condition during the Phase-II inspection.

NOTE: The flight data download must be done within a maximum of 20 FH. If not, the data may be missed due to the recording time of the unit.

(3) The procedure to find an excess to the landing envelope when an overweight landing occurs are detailed on SUBTASK 210-003-A, and the related inspections are:

1. Applicable to the aircraft structure SUBTASK 210-007-A (Phase I inspection for structure), and SUBTASK 210-008-A (Phase II inspection for structure), and SUBTASK 210-009-A (Phase III inspection for structure).

2. Applicable to the engines as per SUBTASK 200-002-A (Phase I inspection for engines), and SUBTASK 200-004-A (Phase II inspection for engines).

- Do the hard landing inspection (AMM TASK 05-50-03-200-801-A/600) if the crew reports one of these two conditions:

1. The aircraft touches the ground with the NLG first.

2. The crew reports a suspect hard landing in the NLG because of high rate of pitch-angle variation.

(4) For an overweight landing analysis, these parameters from the flight data source are necessary (AMM TASK 31-31-00-970-801-A/200 or AMM TASK 31-31-00-970-802-A/200 or AMM TASK 31-32-00-470-801-A/200):

- Vertical CG acceleration (NormAccel)
- Air/Ground indication (Air/Gnd)
- Gross weight (GrossWeight)
- Roll (Roll)
- Time (GMT)
- Date (Day)
- Flight number (FltNumber)
- Pitch angle (Pitch)

NOTE: The data must be selected from the flight data source at a rate of 8 sps (samples per second). A higher rate if used will not be compatible with the monitors thresholds.

(5) The procedures given here let you find an envelope exceedance condition and put the related aircraft in a serviceable condition. If you will not do the Phase-III inspection, it is not necessary to send data to Embraer. But, if it is necessary to do the Phase-III inspection, send the data to Embraer for analysis (flight data source and a report of the damage found). The decision to put the aircraft back into service will then be made by Embraer together with you.

(6) When the conditional inspection tells you to "examine" a component, you must look for the conditions that follow:

- Pulled-apart structure
- Loose paint (paint flakes)
- Discoloration
- Distortion (twisted parts)
- Wrinkles or buckles in the structure
- Bent components
- Fastener holes that become oversized
- Loose fasteners
- Fasteners that pulled out or are missing
- Delamination
- Fiber breakouts
- Black powder residues around, and that trail from, fasteners in composites (indication of loose fasteners)
- Misalignment
- Interference
- Nicks or gouges
- Scratches
- Other signs of damage

(7) All the inspections are classified as GVI.

(8) If during the inspection of the structures there are signs of damage related to the overweight landing event, contact Embraer for the airworthiness restoration of the related aircraft.

E. Overweight Landing Detection

SUBTASK 210-003-A

(1) The check of vertical CG acceleration and roll rate parameters obtained from flight data analysis against their thresholds will classify the landing as a landing event inside the aircraft envelope for which no further action is necessary or a suspect landing event outside of the aircraft envelope when the inspections apply.

(2) Due to complexity of landing loads and their effect on the aircraft structure, the analysis differ between aircraft structure and aircraft engines.

(3) Vertical CG acceleration check for aircraft structure:

(a) In the flight data, mark the initial and the final data samples to specify the range for vertical CG acceleration analysis (Figure 602):

1 The range starts 4 SEC before the first "Ground" recorded for the Air/ Ground parameter.

2 The range ends at the first sample with a value smaller than or equal to -0.5 degrees for the pitch angle, after the transition of the Air/Ground parameter from "Air" to "Ground". Refer to Figure 602.

(b) Get the maximum value of vertical CG acceleration in the selected range and the parameter "Gross Weight" recorded by the DVDR, in the air-to-ground transition.

(c) If the vertical CG acceleration peak recorded by the flight data is not higher than the vertical CG acceleration threshold for the given gross weight value, see Figure 602, no other action is necessary and the aircraft structure is cleared for service.

(d) If the vertical CG acceleration peak recorded by the flight data is higher than the vertical CG acceleration threshold for the given gross weight value, see Figure 602, do the aircraft structure Phase-II inspection.

NOTE: Make sure that you have completed aircraft structure Phase I inspection before starting Phase II inspection.

(e) If you do not find an incorrect condition during the aircraft structure Phase-II inspection, no other action is necessary and the aircraft structure is cleared for service. If you find an incorrect condition during the aircraft structure Phase-II inspection, do the aircraft structure Phase-III inspection and send Embraer these data:

1 Flight data that contain the overweight landing incident.

2 Report of the incorrect conditions that you found during the preceding inspections.

NOTE: Whenever aircraft structure Phase III inspection is required, the engines Phase II inspection is also required.
When you do a Phase-III inspection on an aircraft, you must send Embraer the flight data and a report of the damage that you found. You and Embraer together will then make the decision to put the aircraft back into service.

(4) Vertical CG acceleration check for engines:

(a) The Vertical CG acceleration is the same calculated in SUBTASK 210-003-A.

NOTE: The engines maximum value of vertical CG acceleration threshold does not change with the aircraft weight variation.

(b)

If acceleration threshold of 2.30 G for any aircraft mass during landing be exceeded, do the engines Phase II inspection.

NOTE: Make sure that you have completed engines Phase I inspection before starting Phase II inspection.

(5) Roll Rate Check

(a) NOTE: If the maximum value of vertical CG acceleration from the previous step does not exceed 1.35 G you can skip directly the roll rate check for the overweight landing analysis.

In the flight data, mark the initial and the final data samples to specify the range for roll data for analysis (Figure 603):

1 The range starts 2 SEC before the higher value recorded for the vertical CG acceleration.

2 The range stops 2 SEC after the higher value recorded for the vertical CG acceleration.

(b) Use the selected range of values for roll to calculate the roll rate. Refer to Figure 603.

(c) Get the parameter "Gross Weight" recorded by the DVDR, in the air-to-ground transition.

(d) With the two parameters (Maximum roll rate and Gross Weight), use the chart in Figure 603 to classify the actions accordingly.

NOTE: If aircraft structure phase II inspection is required, make sure that you have completed aircraft structure phase I inspection before starting Phase II inspection.

(e) If you do not find an incorrect condition during the aircraft structure Phase-II inspection, no other action is necessary and the aircraft structure is cleared for service. If you find an incorrect condition during the aircraft structure Phase-II inspection, do the aircraft structure Phase III inspection and send data to Embraer.

NOTE: Whenever aircraft structure Phase III inspection is required, the engines Phase II inspection is also required.
When you do an aircraft structure Phase III inspection on an aircraft, you must send Embraer the flight data and a report of the damage that you found. Customer and Embraer together will then make the decision to put the aircraft back into service.

(6) Roll rate check for engines

(a) The Roll rate is the same calculated in SUBTASK 210-003-A.

NOTE: The engines maximum value of roll rate threshold does not change with the aircraft weight variation.

(b) If the roll rate threshold of 13.0 dg/sec for any aircraft mass during landing be exceeded, do the engines Phase II inspection.

NOTE: Make sure that you have completed engines Phase I inspection before starting engines Phase II inspection

F. Job Set-Up

SUBTASK 940-002-A

(1) Do the procedure to make the aircraft safe for maintenance (AMM TASK 20-00-00-910-801-A/200).

SUBTASK 940-003-A

(2) Put all flight control surfaces in the neutral position.

G. Phase-I Inspection for aircraft structure

SUBTASK 210-007-A

(1) Do a visual inspection of the items shown below. Do not disassemble them.

(2) Main Landing Gear

(a) Examine the MLG wheels for distortion, bending, dents and fracture.

(b) Examine the MLG tires for general condition and deflation.

(c) Examine the brake assembly for damage, and the hoses for signs of fluid leakage.

(d) Examine the MLG wheelwell for signs of fluid leaks.

(e) Examine the MLG locking stay for distortion, bending and fracture.

(f) Examine the MLG torque link for distortion, bending and fracture.

(g) Examine the upper and lower MLG side stay for distortion, bending and fracture.

(h) Examine the MLG strut for leakage around the sliding tube diameter.

(3) Fuselage

(a) Examine the external surface of the fuselage for fluid leakage.

(b) Examine the external areas near the wing stub for any signs of damage.

(c) Open and close the forward and aft passenger doors. Make sure that they operate correctly (AMM SDS 52-11-00/1 and AMM SDS 52-12-00/1).

(d) Open and close the forward and aft service doors. Make sure that they operate correctly (AMM SDS 52-41-00/1 and AMM SDS 52-42-00/1).

(e) Open and close the forward and aft cargo doors. Make sure that they operate correctly (AMM SDS 52-31-00/1 and AMM SDS 52-32-00/1).

(f) Examine the tail cone and silencer.

(g) Examine the wing-to-fuselage fairing junction with the fuselage and look for any missing or damaged screw, missing sealant, and paint peel-off.

(4) Wing

(a) Examine the MLG doors.

(b) Examine the external surface of the wing for fuel or other fluid leakage.

NOTE: You must carefully examine the fuel-tank access-panel contours for fuel leakage.

(c) Examine the pylon fairings for signs damage.

(5) Flight Controls

(a) Examine all flight controls to make sure that they move freely.

(b) With the flaps fully retracted, examine the flap surfaces for any misalignment.

NOTE: If flap rods are bent but no other related parts are damaged, only flap rod replacement is necessary. In this case, if you don't have any other discrepancy during phase I inspection, phase II inspection is not necessary.

H. Phase-I Inspection for aircraft engines

SUBTASK 200-002-A

(1) Engines

(a) Examine the inlet-cowl internal-skin for signs of chafing with the fan blades.

I. Phase-II Inspection for aircraft structure

SUBTASK 210-008-A

(1) Main Landing Gear

(a) Examine the area above the MLG actuator support-attachment for buckled skin.

(b) Do a check of the gaps of the proximity sensors. The sensor surfaces must be parallel.

(c) Examine the MLG locking-stay pins for distortion, bending and fracture.

(d) Examine the MLG torque link pins for distortion, bending and fracture.

(e) Examine the upper and lower MLG side-stay pins for distortion, bending and fracture.

(f) Examine the MLG main fitting pintle pins for distortion, bending and fracture.

(g) Examine the trunnion fitting attachment.

(h) Examine the wheel axle to know if it is bent.

(i) If a BRK OVERHEAT message shows or if one or more wheel fuse plugs were released, you must do a high-energy stop inspection (AMM TASK 05-50-12-200-801-A/600).

(j) Examine the MLG shimmy damper (AMM TASK 32-11-03-200-802-A/600).

(k) Do a check of the MLG pintle pins for distortion.

(2) Fuselage

(a) Externally examine the upper and lower fuselage-skin panels for loose or sheared rivets, structural damage, and signs of fuel and other fluid leakage.

NOTE:
- For the inspection of the lower skin panels, do one of these two options:
  1- Remove the wing-to-fuselage fairings (AMM MPP 53-04-20/400, AMM MPP 53-04-30/400, AMM MPP 53-04-40/400, and AMM MPP 53-04-50/400).
- 2- Remove access panels 191AR, 191EL, 191BL, and 191CL and a borescope inspection should be done.
- If you find wrinkles in the skin panels, you must do an internal visual inspection on the fuselage.

(b) Carefully examine all the systems in the wing-to-fuselage fairing region and its attachment to the fuselage.

(c) On the forward avionics compartment, examine the MAU/EICC rack and shelves for integrity.

(d) On the middle avionics compartment, examine the avionics rack and shelves for integrity.

(e) On the rear fuselage, examine the avionics racks and shelves for integrity.

(f) Carefully examine the internal side of the fuselage in the attachment of the galley to the frame. Refer to Figure 604, Sheet 1, Figure 604, Sheet 2, and/or Figure 604, Sheet 3.

NOTE: It is not necessary to remove the galley to get access to the inspection area.

(3) Wing

(a) Examine wing spars II and III. Start from the MLG wheelwell. Remove the access panels and, with the inboard flap in the down position, also examine the lower skin adjacent to them.

(b) Examine the upper and lower wing skin panels from the wing root to the wing tip.

(c) Externally examine the wing, nacelles, and engines struts for loose or sheared rivets, structural damage, and signs of fuel and other fluid leakage.

(d) Examine the wing-to-fuselage junction and the MLG wheelwell for damage. Look for flaked paint and loose or missing fasteners.

(4) Flight Controls

(a) With the flaps fully deflected, examine the flap and slat mechanism for bent components.

(b) With the flaps fully retracted, examine the flap and slat surfaces for any misalignment.

NOTE: If flap rods are bent but no other related parts are damaged, only flap rod replacement is necessary. In this case, if you don't have any other discrepancy during phase II inspection, phase III inspection is not necessary.

(c) In the MLG compartment, do a check to see if the aileron cable and the pulleys are in the correct position.

J. Phase-II Inspection for aircraft engines

SUBTASK 200-004-A

(1) Engines

(a) Do the GE engines Manual over-limit condition Inspection LMM SUBTASK 72-00-00-810-027.

K. Phase-III Inspection for aircraft structure and aircraft engines

SUBTASK 210-009-A

(1) Defuel the aircraft (AMM TASK 12-11-01-650-802-A/300).

(2) Put the aircraft on jacks (AMM TASK 07-10-01-500-801-A/200).

(3) Examine the landing gear for correct operation of the free-fall system (AMM TASK 32-34-00-710-801-A/500).

NOTE: An irregular, not-smooth operation of the landing gear during free-fall can show deformation of the landing gear. You must then examine it for correct alignment.

(4) Carefully examine the internal side of the CF II in the area below the floor cross beams and columns.

(5) Do the aircraft alignment check (AMM TASK 51-50-00-800-801-B/600).

L. Job Close-Up

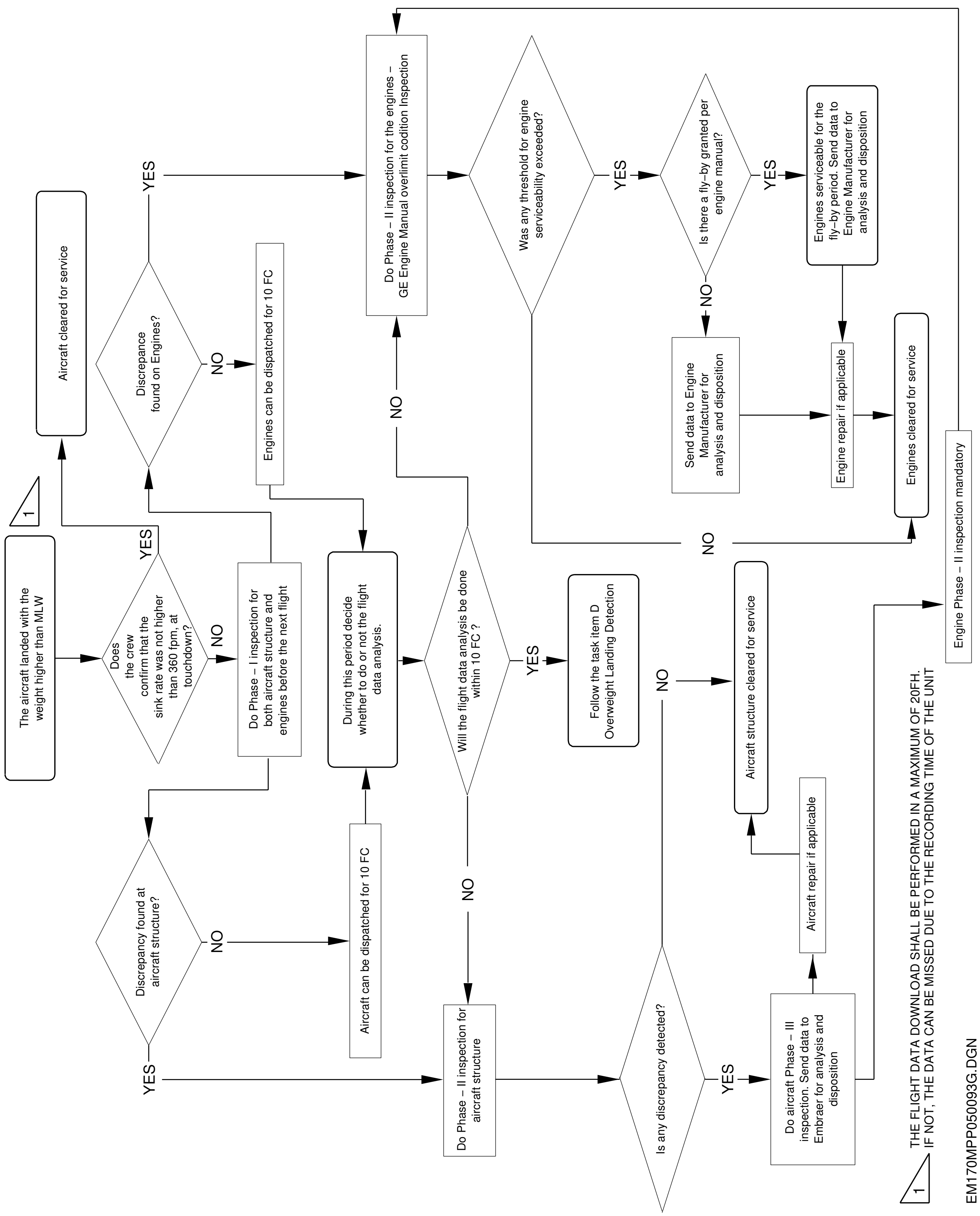
SUBTASK 940-004-A

(1) Put the aircraft back to its initial condition (AMM TASK 20-00-00-910-801-A/200).

EFFECTIVITY: ALL

Overweight-Landing Inspection Flowchart

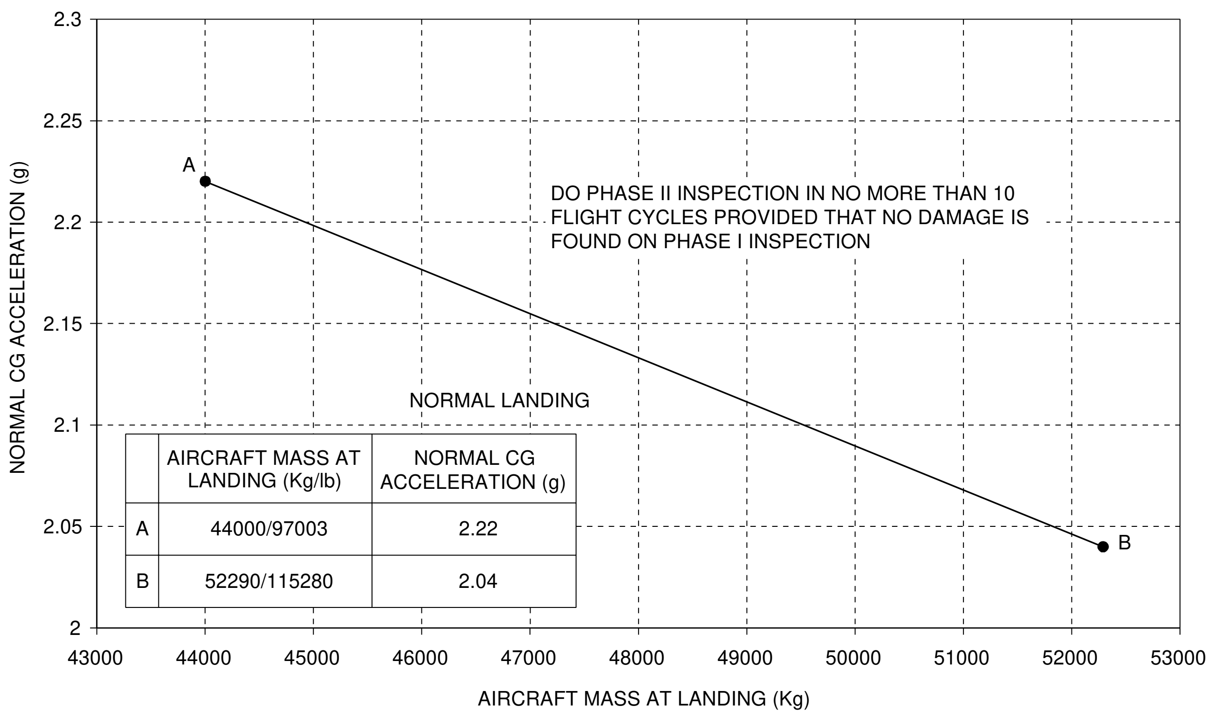
Figure 601



EFFECTIVITY: ALL

NZ Threshold

Figure 602



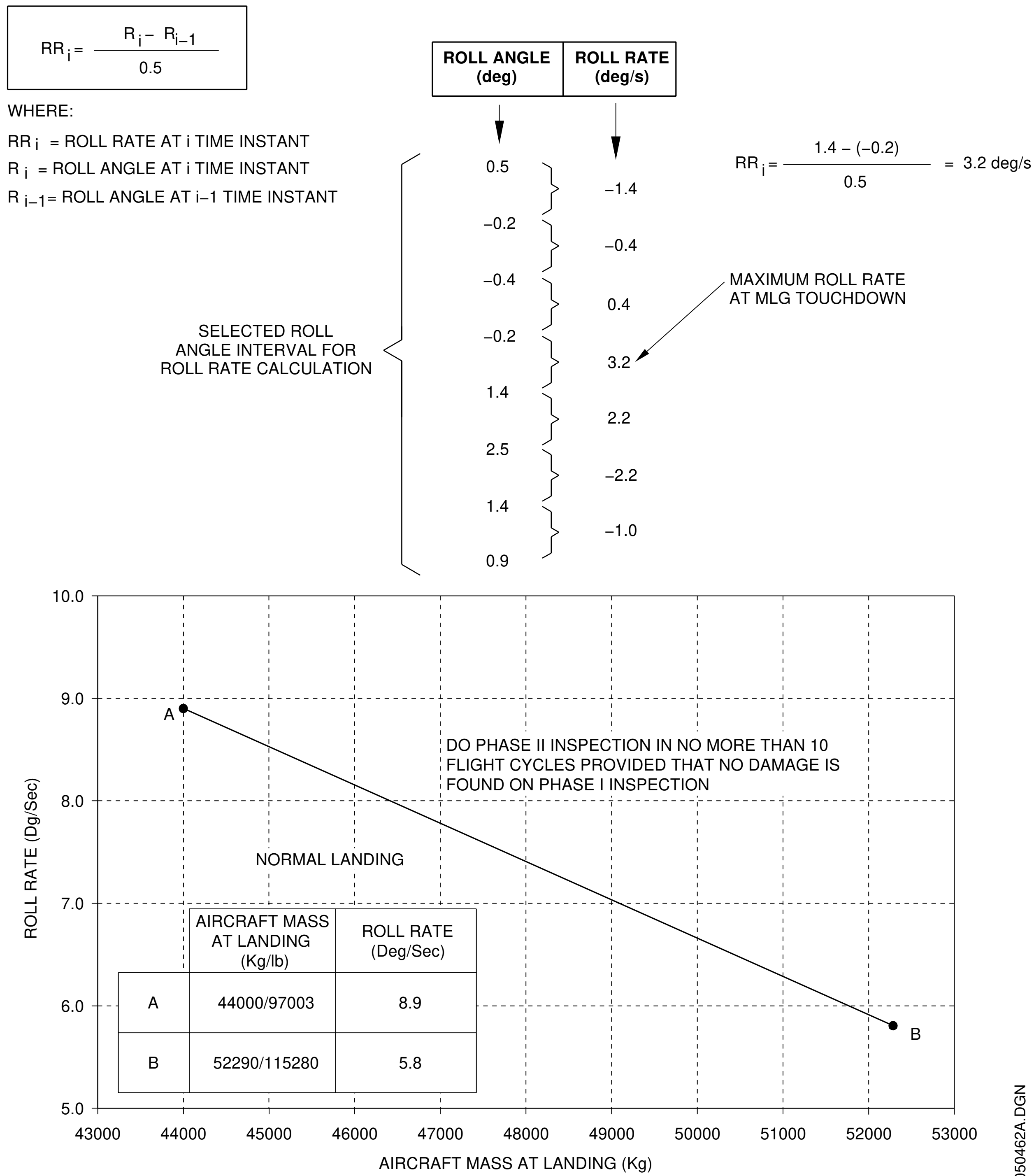
	AIRCRAFT MASS AT LANDING (Kg/lb)	NORMAL CG ACCELERATION (g)
A	44000/97003	2.22
B	52290/115280	2.04

EM170MPP050461A.DGN

EFFECTIVITY: ALL

Roll Rate Calculation and Threshold

Figure 603



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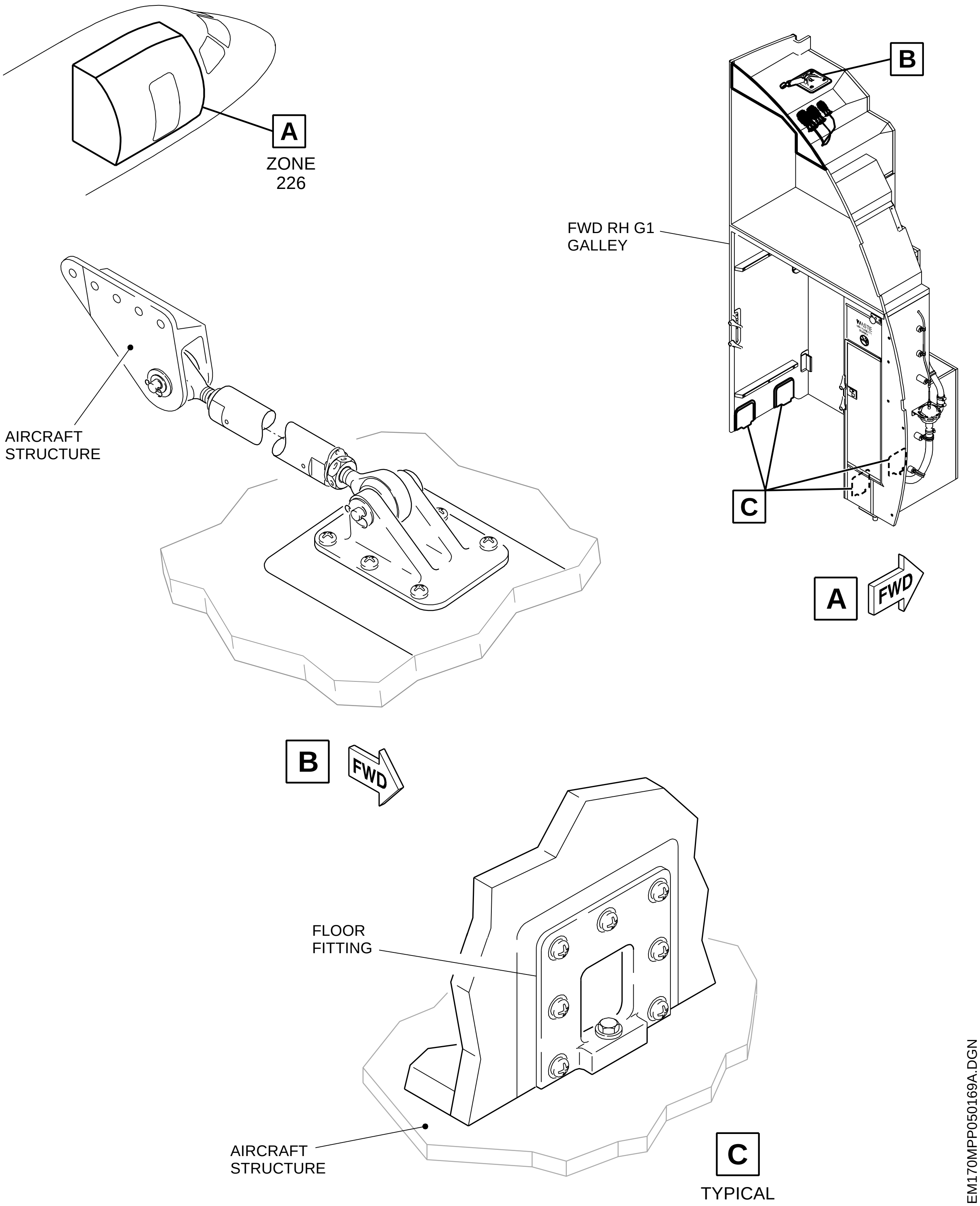
NOTES:

IN CASE THE MAXIMUM ROLL RATE VALUE IS NEGATIVE, USE THE ABSOLUTE VALUE.

SKIP ROLL RATE ANALYSIS IF MAXIMUM VERTICAL ACCELERATION IS LESS THAN 1.35 G

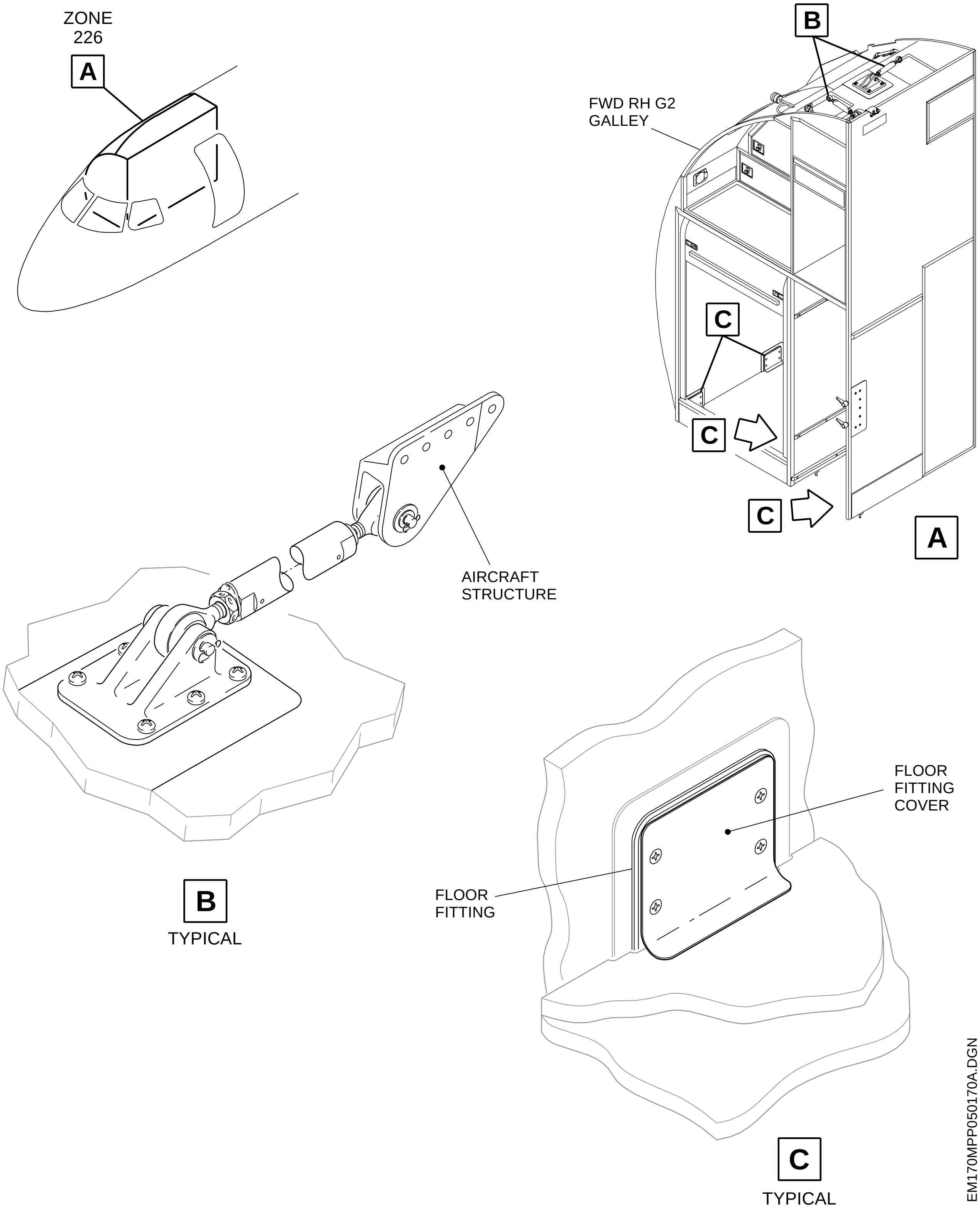
EFFECTIVITY: ALL
Galley Attachments - Inspection
Figure 604 - Sheet 1



EFFECTIVITY: ALL

Galley Attachments - Inspection

Figure 604 - Sheet 2



EFFECTIVITY: ALL
Galley Attachments - Inspection
Figure 604 - Sheet 3